

CM0081 Formal Languages and Automata

§ 3.4 Algebraic Laws for Regular Expressions

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Preliminaries

Conventions

- The number and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook (Hopcroft, Motwani and Ullman 2007).
- The natural numbers include the zero, that is, $\mathbb{N} = \{0, 1, 2, \dots\}$.
- The power set of a set A , that is, the set of its subsets, is denoted by $\mathcal{P}A$.

Algebraic Laws for Regular Expressions

Definition

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Let M , N and P be regular expression variables.

Sugar syntax

$$M^+ := MM^*,$$

$$M? := \varepsilon + M.$$

Algebraic Laws for Regular Expressions

Some laws for union

$$(M + N) + P = M + (N + P)$$

(associativity)

$$M + \emptyset = \emptyset + M = M$$

(identity)

$$M + N = N + M$$

(commutativity)

$$M + M = M$$

(idempotence)

Observation

There is no inverse for union.

Algebraic Laws for Regular Expressions

Some laws for concatenation

$$(MN)P = M(NP)$$

(associativity)

$$M\varepsilon = \varepsilon M = M$$

(identity)

$$MN \neq NM$$

(non-commutativity)

$$M\emptyset = \emptyset M = \emptyset$$

($\emptyset$ is the annihilator for concatenation)

Observation

There is no inverse for concatenation.

Algebraic Laws for Regular Expressions

Some laws for union and concatenation

$$M(N + P) = MN + MP \quad \text{(distributive)}$$

$$(M + N)P = MP + NP \quad \text{(distributive)}$$

Algebraic Laws for Regular Expressions

Some laws for closure

$$(M^*)^* = M^*$$

(idempotence)

$$\emptyset^* = \varepsilon$$

$$\varepsilon^* = \varepsilon$$

$$(\varepsilon + M)^* = M^*$$

$$M^* = M^+ + \varepsilon$$

Algebraic Laws for Regular Expressions

Observation

A complete set of axioms for the regular expressions is indicated in (Kozen 2012, Lecture 9).

Simplification of Regular Expressions

Example

$$\begin{aligned}0 + (\varepsilon + 1)\underline{(\varepsilon + 1)^*}0 &= 0 + \underline{(\varepsilon + 1)1^*}0 \\&= 0 + (\underline{\varepsilon 1^*} + 11^*)0 \\&= 0 + (1^* + \underline{11^*})0 \\&= 0 + (\underline{1^*} + 1^+)0 \\&= \underline{0 + 1^*}0 \\&= 1^*0\end{aligned}$$

$$((\varepsilon + M)^* = M^*)$$

(distributive)

(identity)

(def. L^+)

(equivalence)

(equivalence)

Discovering Laws for Regular Expressions

Method

Let E and F be two regular expressions with the same set of variables $\{M_1, \dots, M_n\}$.

To test if $E = F$:

1. Convert E and F to **concrete** regular expressions C and D , replacing each M_i by a different symbol a_i , for $i = 1, 2, \dots, n$.
2. Test whether $L(C) = L(D)$. If so, then $E = F$, and if not $E \neq F$.

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Observation

We are proving by example!

Discovering Laws for Regular Expressions

Example

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We replace the variable M by the concrete regular expression a .

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Because $L(a^*) = L(a^*a^*)$ we conclude that

$$M^* = M^*M^*.$$



Discovering Laws for Regular Expressions

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Refutation

We replace the variables M and N by the concrete regular expressions a and b respectively.

$$a + ba \stackrel{?}{=} (a + b)a.$$

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We replace the variables M and N by the concrete regular expressions a and b respectively.

$$a + ba \stackrel{?}{=} (a + b)a.$$

Because $aa \notin L(a + ba)$ and $aa \in L((a + b)a)$ then

$$L(a + ba) \neq L((a + b)a).$$

So, we can conclude

$$M + NM \neq (M + N)M.$$



Discovering Laws for Regular Expressions

Example (Exercise 3.4.2.d)

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Discovering Laws for Regular Expressions

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Prove or disprove that $(M + N)^*N = (M^*N)^*$.

Refutation

We replace the variables M and N by the concrete regular expressions a and b respectively.

$$(a + b)^*b \stackrel{?}{=} (a^*b)^*.$$

Since $\varepsilon \notin (a + b)^*b$ and $\varepsilon \in (a^*b)^*$ then

$$(M + N)^*N \neq (M^*N)^*.$$



Discovering Laws for Regular Expressions

Example (counter-example)

Extensions of the previous test beyond regular expressions may fail.

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1. Add $\cap$ to the regular expression operators.
2. Test if $M \cap N \cap P = M \cap N$.
3. From $M = a$, $N = b$ and $P = c$ and since

$$\{a\} \cap \{b\} \cap \{c\} = \emptyset = \{a\} \cap \{b\},$$

we should conclude that the “property” is true.

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we should conclude that the “property” is true.

4. But, the “property” is false. For example, if $M = N = a$ and $P = \emptyset$ then

$$M \cap N \cap P \neq M \cap N.$$

5. Therefore, the test is not valid!

References

-  John E. Hopcroft, Rajeev Motwani and Jefferey D. Ullman [1979] (2007). Introduction to Automata Theory, Languages, and Computation. 3rd ed. Pearson Education (cit. on p. 2).
-  Dexter C. Kozen [1997] (2012). Automata and Computability. Third printing. Undergraduate Texts in Computer Science. Springer. DOI: [10.1007/978-1-4612-1844-9](https://doi.org/10.1007/978-1-4612-1844-9) (cit. on p. 10).